01 / DESIGN

New Solar Sample

220.4 kWp rooftop solar system

Client Mr. Jaisawal | Site 36.1691° N, 115.1461° W | Date 14 Oct 2026



SYSTEM SIZE	MODULES	ROOF AREA USED	ANNUAL GENERATION
220.4 kWp	380 × 580 W	3,749 m²	353.8 MWh
		37% of the roof	year 1 estimate

Explore your design in 3D

Scan to open the live 3D model of your roof — walk around it, check every panel and watch the sun move across the day.

Open 3D view →

02 / SHADOW

Shadow analysis

How much sunlight reaches the panels through the year

98.7%
solar access

Excellent solar access

SHADING LOSS	PEAK SHADING	SUN POSITIONS
1.3 %	5:00 PM	4
of yearly energy	18% of panels affected	checked through the year









Assessment. Across the representative sun positions through the year, the proposed layout sees minimal shading — about 1.3% of yearly energy.

03 / PERFORMANCE

System performance

Expected energy and where it is lost

ANNUAL GENERATION · YEAR 1

353.8 MWh

≈ 29.5 MWh a month on average

SPECIFIC YIELD	PERFORMANCE RATIO	SYSTEM SIZE
1,605 kWh/kWp	78.4 %	220.4 kWp

System losses

largest first · share of ideal energy

Temperature		10.3%
Soiling		3.0%
Inverter		3.0%
Mismatch		2.0%
DC Wiring		2.0%
Shading		1.3%
Total	combined effect of all losses	**21.6%**

System summary

SYSTEM CAPACITY **220.4 kWp**	MODULES **380 × 580 W**	ROOF AREA USED **3,749 m²**
INVERTERS **2 × 50 kW**	DC / AC RATIO **2.20**	TILT / AZIMUTH **27° / 207° (SW)**

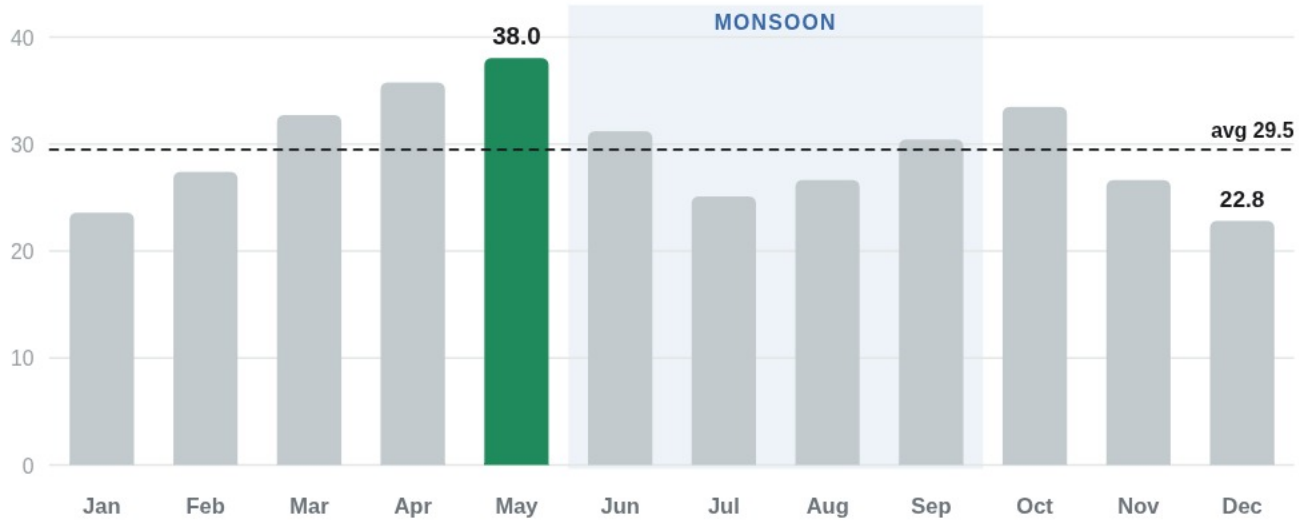
04 / GENERATION

Annual generation

12-month energy forecast

YEAR 1 TOTAL	BEST MONTH	LOWEST MONTH	MONTHLY AVERAGE
353.8 MWh	May 38.0 MWh	Dec 22.8 MWh	29.5 MWh

Monthly generation (MWh)



Jan	Feb	Mar	Apr
23.6 MWh	27.4 MWh	32.7 MWh	35.8 MWh
107 kWh/kWp · 6.7%	124 kWh/kWp · 7.7%	148 kWh/kWp · 9.2%	162 kWh/kWp · 10.1%

May BEST	Jun MONSOON	Jul MONSOON	Aug MONSOON
38.0 MWh	31.2 MWh	25.1 MWh	26.6 MWh
173 kWh/kWp · 10.8%	142 kWh/kWp · 8.8%	114 kWh/kWp · 7.1%	121 kWh/kWp · 7.5%

Sep MONSOON	Oct	Nov	Dec
30.4 MWh	33.5 MWh	26.6 MWh	22.8 MWh
138 kWh/kWp · 8.6%	152 kWh/kWp · 9.5%	121 kWh/kWp · 7.5%	104 kWh/kWp · 6.5%